Anomaly Detection

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Machine Learning

Broader Themes

- Problem formulation
- Feature construction
- Threshold selection
- Multi-view learning
- From unsupervised → supervised

What is Anomaly Detection?

- **Goal**
  - Identify instances that deviate from expected behavior
- **Key Challenge**
  - We usually do **NOT** have labeled anomalies

Three Types of Pattern Discovery

Goal	Knowledge Level	Example
Well-defined patterns	We know what to look for	Pattern matching
Vaguely-defined patterns	**General expectation**	**Anomaly detection**
Undefined patterns	No expectations	Pattern mining

Real-World Applications

- **Insider threat detection**
- Health monitoring
- Fraud detection
- Intrusion detection
- Fault detection in systems

Insider Threat Detection

- **Problem**
    - Identify suspicious employees based on computer activity
- **Available Data**
    - Logon activity
    - File access
    - Email
    - Web browsing
    - Device usage
- **We do NOT know:**
    - What is malicious
    - True job roles (privacy constraints)



Detection Design Considerations

- **Cost of Errors**

Error Type	Definition	Example Cost
False Positive	Flag benign as suspicious	Investigation cost
False Negative	Miss true anomaly	Financial loss / human harm

- **Key Insight**
  - *Threshold choice reflects cost trade-offs*

From Raw Data → Features

- **Challenge:** High-dimensional, multi-modal activity logs
- **Solution:** Feature engineering per domain
- **Examples:**
  - **Logon:** unique machines, After-hours frequency
  - **Email:** % external emails, Mass email frequency
  - **Web:** Topic distribution
- **Approach:** *Raw Logs → Feature Extraction → Domain-specific features*



From Features → Behavior Modeling

- **Problem:** Multi-source data with weak connections
- **Approach:** Treat each activity type as a separate domain
- **Benefits:**
  - Reduces noise
  - Improves interpretability
  - Enables multi-view modeling

What is “Suspicious”?

- We consider two definitions:
  - Unusual change over time
  - Blend-in anomaly



Unusual Change Detection

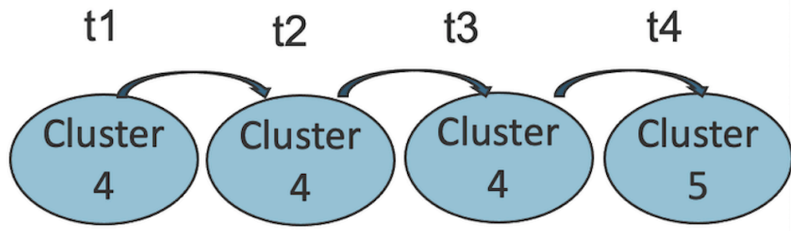
Change Detection

- **Idea:** If a user's behavior distribution changes → suspicious
- **Example:** Web topics suddenly change
- **Limitations:**
  - Assumes stable behavior
    - Counterexample: Bob joins a new project → legitimate shift
  - *High false positive rate*

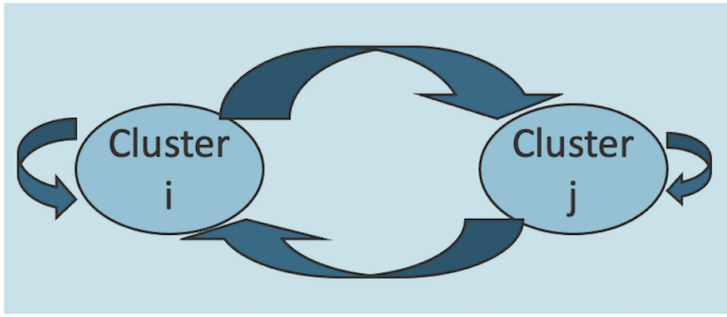
Improved: Peer-Aware Change Detection

- **Idea:** If change is common among peers → not suspicious
- **New Limitation:** Peers may change at different times
- **Solution:** Relax time constraints
- **New Definition:** Rarely observed change → suspicious
- *“Suspicious = Unusual Change”*

Modeling Unusual Change



- **Step 1:** Cluster users daily



- **Step 2:** Model as Markov chain over clusters

  c_t^u = cluster of user u at time t

  $p(c_0)$ = prior

  $q(c_i, c_j)$ = transition probability

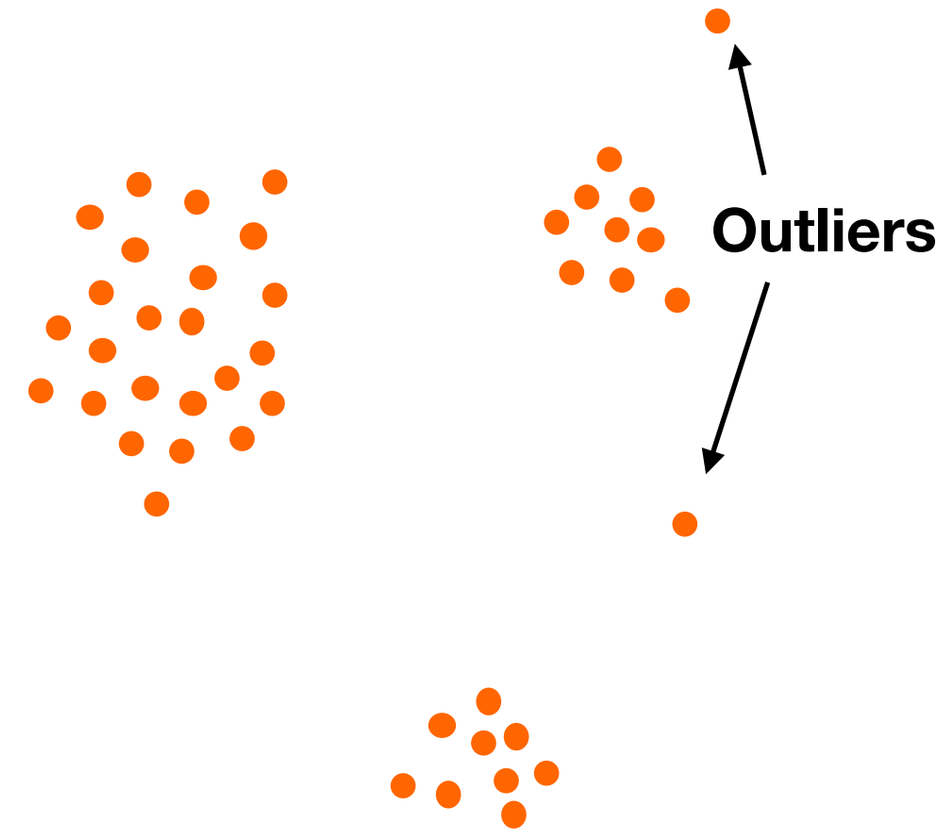
- **Suspicion Score** $S_u = p(c_0^u) \prod_{t=1}^{n-1} q(c_t^u, c_{t+1}^u)$

- *Low probability sequence ⇒ high suspicion*

Blend-in Anomalies

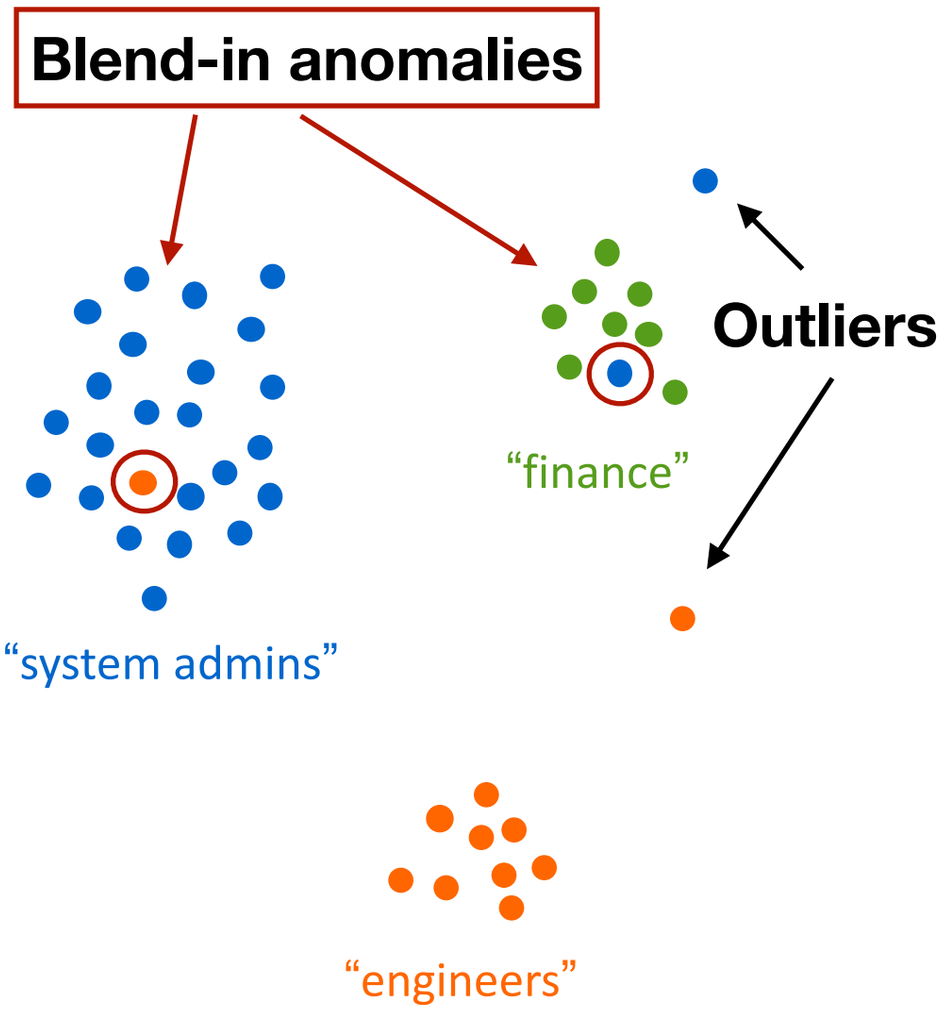
Why Outlier Detection Alone Fails

- **Classic method:**
  - Cluster users
  - Detect outliers
- **Problem: Blend-in Anomalies**



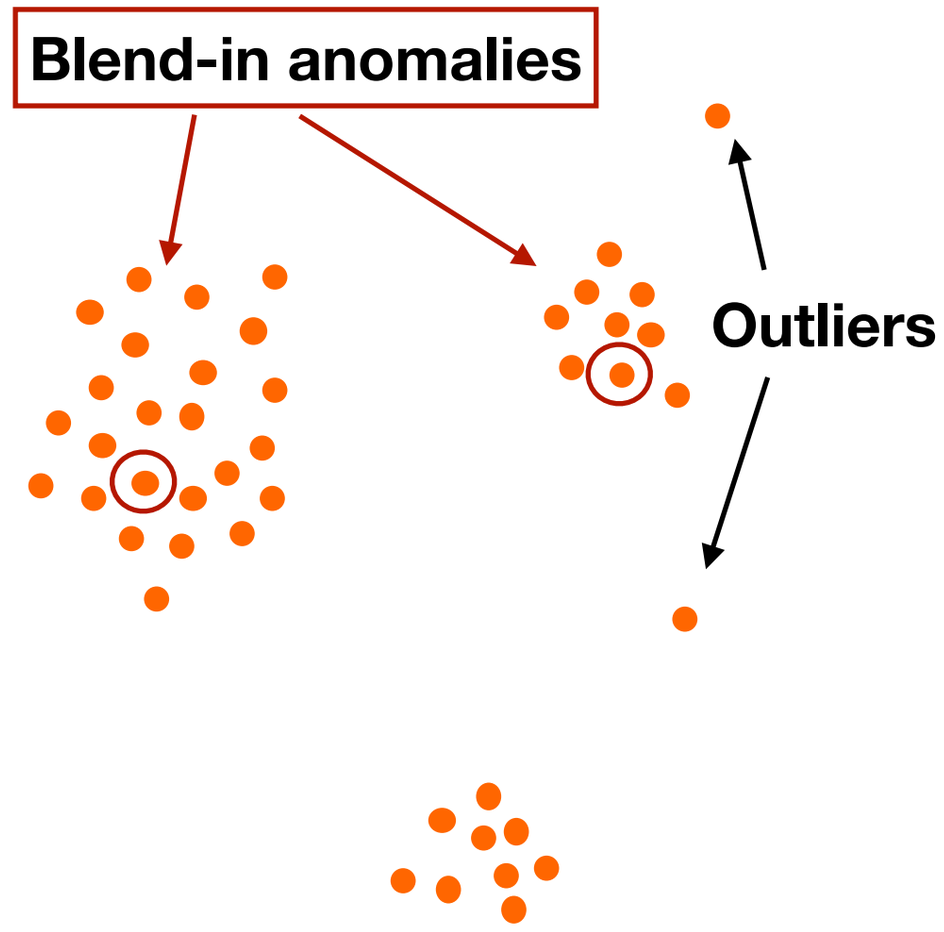
Blend-in Anomalies

- **False Negative Example:**
  - Eve (engineer) behaves like System Admin
  - She is:
    - Not an outlier
    - But behaving outside her role

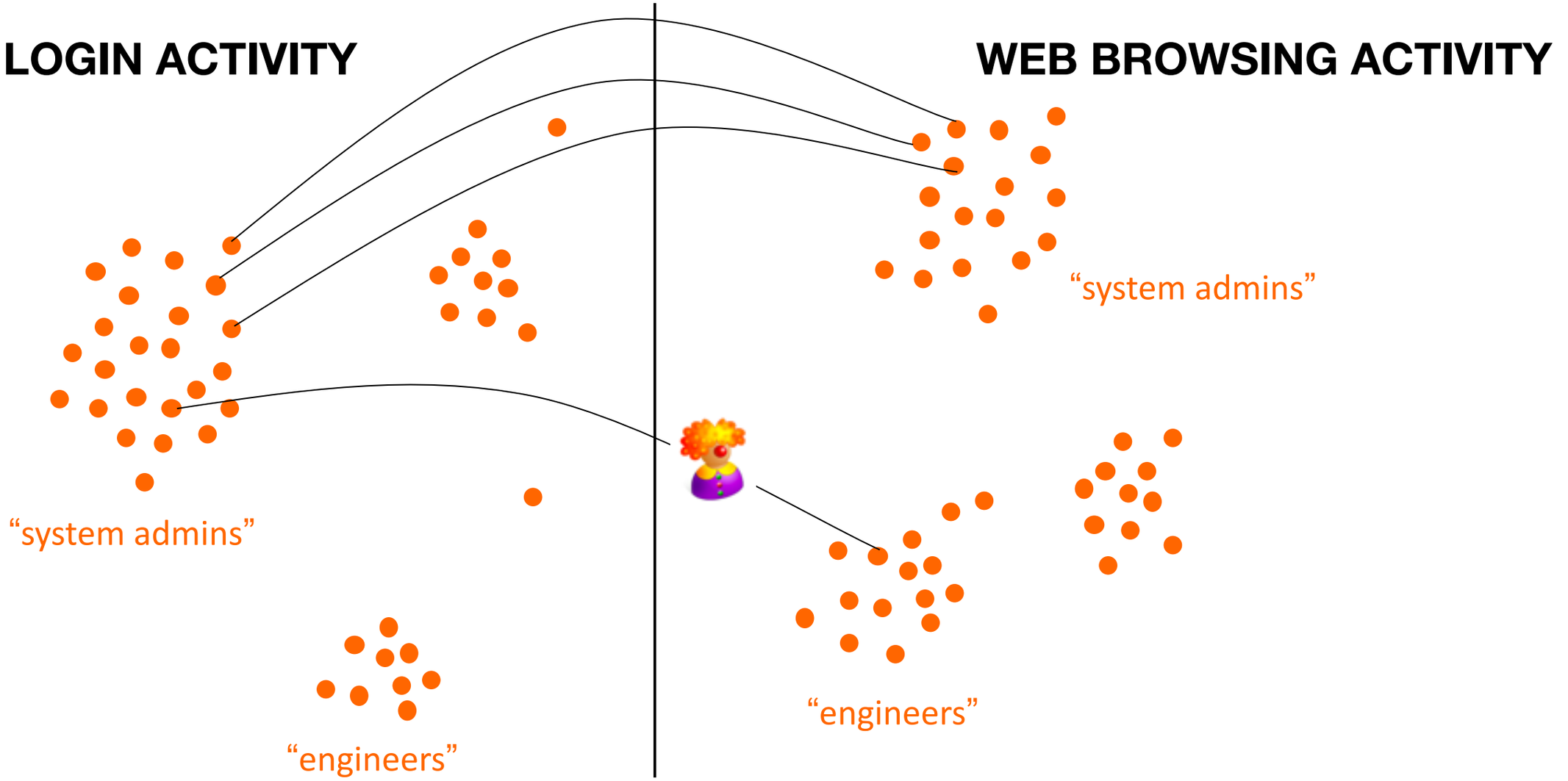


Blend-in Anomaly Detection

- **Challenge:**
  - Job roles (cluster labels) hidden for privacy
  - *Outliers blend in*



Formalizing Blend-in Anomalies



Definition: A user behaves consistently within a domain but inconsistently across domains

Cross-domain inconsistency ⇒ suspicious

Key Insight: Multi-Domain Consistency

- **Assumption:**
  - Benign users have predictable peer groups across domains
- **Therefore:**
  - We can predict peer group in one domain from peer groups in other domains

Turning Unsupervised into Supervised

- Step 1: Cluster each domain independently
- Step 2: Use clusters from other domains to predict cluster in target domain
- Step 3: Users with low prediction probability are suspicious

domain	Logon	Device	File	Email	Web browsing
User 1	1	3	4	2	
User 2	1	2	3	1	
User 3	2	2	3	2	
User 4	1	2	3	1	
User 5	1	2	3	1	
User …	1	2	3	1	
User …	1	2	3	2	?

- Suspicion (Blend-In) Score: $P(c^d \mid c^{\neg d})$
- Low predictability ⇒ high anomaly score

Combining Evidence

Multi-Domain Fusion

- We now have multiple anomaly scores per user:
  - Change score (Logon)
  - Change score (Email)
  - Blend-in score (File)
  - Blend-in score (Web)
- *How do we combine them?*

Fusion Levels

- **Model-Level Fusion**
  - Cross-domain constraints inside model
- **Score-Level Fusion**
  - Combine anomaly scores (sum, weighted sum)
- **Rank-Level Fusion**
  - Combine user rankings across domains
- *“Domain models → Scores → Ranking”*

Work practice data

Logon

File

Device

Email

Http

Feature extraction

Clustering

Feature extraction

Clustering

Feature extraction

Clustering

Feature extraction

Clustering

Feature extraction

Clustering

to discover peer groups

Model-level Fusion

Multi-domain cross validation

Across domain

Across time

Unusual change detection

Information fusion

Suspect list

Score-Level Fusion

Goal: combine suspicion scores from multiple activity domains

<u>Each domain provides a different amount of information</u>

Weight scores based on inverse frequency of suspicious behavior

$$w_d = \log \frac{N}{M_d}$$

w: weight, d: domain, N: #users, $M_{d:}$ #suspicious users in domain d

$$c_d^u = \sum_d w_d * s_d^u$$

c: combined score, s: score, u: user

Ranking-based Fusion

U	d1	d2	d3	d4	Comb	MinRank	#Indicators
1	**4**	**7**	**1**	**9**	18	1	4
2	**8**	47	**6**	**1**	17	1	3
3	**1**	**6**	**3**	28	50	1	3
4	15	**9**	**8**	**2**	20	2	3
5	11	**2**	**7**	99	11	2	2
6	**2**	12	**4**	77	72	2	2
7	27	**3**	31	16	38	3	1

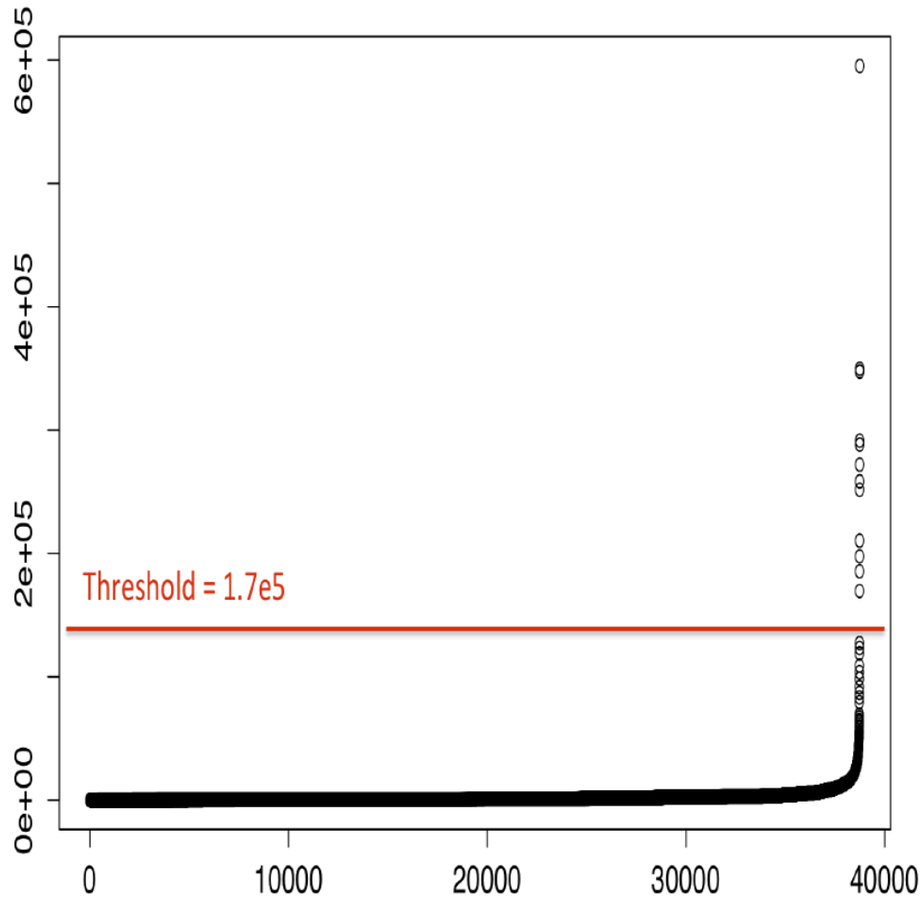
- *Robust to scale differences*

Identifying Outliers in Scores

- Given score set S:
  - **Naive:**
    - Top-k
    - Limitation: May not be true outliers
  - **Statistical:**
    - Score > 3σ
    - Limitation: Assumes Gaussian
  - **Entropy-Based**
    - Measure how much a point increases distribution entropy
    - More flexible

Entropy-based Outlier Identification

- **Goal: identify outliers in a given set of scores S**
- $Entropy\ H(S) = -\sum_{s \in S} P(s) * \log P(s)$
- $Surprise_s = -\log P(s)$
- $SurpriseRatio_s = \frac{surprise_s}{H(S)}$
- $s_{threshold} = s | surpriseRatio_s > 2 * E(surprise_s)$
- $Outliers(S) = \{s \in S | s > s_{threshold}\}$



Summary

- **Problem Formulation**
    - What does “suspicious” mean?
        - Unusual change
        - Blend-in anomaly
- **Detection Methods**
    - Markov change detection
    - Cross-domain peer prediction
- **Fusion Strategies**
    - Model-level
    - Score-level
    - Rank-level

Final Takeaways

- **Anomaly detection is difficult because:**
    - No labels
    - High-dimensional data
    - Ambiguous definitions
    - Cost-sensitive decisions
- **But powerful when:**
    - Combined across domains
    - Thoughtfully modeled
    - Carefully thresholded

Related Topics

- Clustering
- Markov models
- Multi-view learning
- Threshold selection